

Multiple integrals – 1

Fubini, the order of integration and symmetry

Homework

13 сентября 2026 года

Theoretical minimum. For non-negative functions the order of integration never matters; in general it does not matter as soon as the integral of the absolute value is finite, and otherwise it may matter a lot. The layer cake formula turns the integral of $|f|^p$ into an integral of the measures of the level sets, and is itself Fubini for the indicator of the subgraph. On a symmetric domain the integrand may be replaced by its symmetrisation, which is the standard first line of a solution. Representing a number as a double integral and then changing the order proves identities between sums and integrals; the same device gives Cauchy's formula for repeated integration and the Dirichlet formula on a simplex. Finally, exchanging a sum and an integral is Fubini on $\mathbb{N} \times \mathbb{R}$, so the counterexamples are of exactly the same kind as for double series.

1. Construct a function on $[0, 1]^2$ for which both iterated integrals exist and are finite but different, and compare with the double series counterexample of Series and infinite products – 5.
2. Determine for which real s the integral $\int_0^1 \int_0^1 |x - y|^{-s} dx dy$ converges, and compute it for those s .
3. Prove that $\iint_{[0,1]^2} (f(x) - f(y))(g(x) - g(y)) dx dy \geq 0$ for monotone f, g of the same direction, and deduce Chebyshev's integral inequality $\int_0^1 fg \geq \int_0^1 f \int_0^1 g$.
4. Prove that $\sum_{n \geq 1} \int_0^1 f_n = \int_0^1 \sum_{n \geq 1} f_n$ for non-negative measurable f_n , and give an example with f_n of variable sign where the two sides differ.

5. Compute $\int_0^\infty \int_0^\infty e^{-xy} \sin x \, dx \, dy$ in two orders and deduce the value of $\int_0^\infty \frac{\sin x}{x} dx$; compare with the two other proofs, in Integrals – 3 and in Improper integrals – 3.
6. Compute $\iint_{[0,1]^2} \ln |x-y| \, dx \, dy$ and $\iint_{[0,1]^2} \max(x, y) \, dx \, dy$, and evaluate $\iint_{[0,1]^2} \lfloor x/y \rfloor \, dx \, dy$.
7. Let f be non-negative and integrable on $\mathbb{R}$. Prove that $\int_{\mathbb{R}} f(x)^2 dx = 2 \iint_{\{f(x) > t > 0\}} dx \, dt$ and deduce the layer cake formula for the square; then use it to prove that $\int f^2 < \infty$ and $\int f < \infty$ do not follow from each other.
8. Prove the Dirichlet formula $\int \cdots \int_{x_i \geq 0, \sum x_i \leq 1} x_1^{a_1-1} \cdots x_n^{a_n-1} dx = \frac{\Gamma(a_1) \cdots \Gamma(a_n)}{\Gamma(a_1 + \cdots + a_n + 1)}$ by induction and Fubini, and deduce the volume of the simplex.
9. Let f be continuous on $[0, 1]$. Prove that $\int_0^1 \int_0^1 \cdots \int_0^1 f(\max(x_1, \dots, x_n)) dx_1 \cdots dx_n = n \int_0^1 t^{n-1} f(t) \, dt$, and find the analogous formula for the minimum and for the median.
10. Prove that for a continuous f on $[0, 1]^2$ the function $F(x) = \int_0^1 f(x, y) \, dy$ is continuous, and construct a bounded f on $[0, 1]^2$ for which both iterated integrals exist but f is not integrable on the square.